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Abstract
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KI:EO – Innovative Ways of Earth Observation for Schools – Use of Artificial Intelligence to Promote Remote Sensing Data in the Classroom

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Challenge

Earth observation (EO) data offer great potential for addressing global issues in geography education and make it possible to monitor environmental processes such as glacier melt or rainforest deforestation in near real time. By comparing satellite imagery from different points in time, these processes become visible and quantifiable. Using EO data, students can better understand abstract concepts such as climate change and environmental destruction and discuss them critically.

However, analysing EO data is challenging for students, and they need additional support that activates deep cognitive structures, for example by scaffolding interpretive reasoning, guiding the translation of complex EO visualisations into conceptual understanding, and prompting learners to engage analytically beyond surface-level pattern recognition. For the latter, teachers need specific knowledge, which many of them lack. Artificial Intelligence (AI), particularly in the form of AI chatbots, holds significant potential to support students and reduce teachers' workloads.

Initial tests with existing chatbots demonstrate that current AI systems are already capable of recognising, locating, and describing EO data. For example, representations of the Normalised Difference Vegetation Index (NDVI) were correctly identified, including explanations of the index and its colour coding. Initial chatbot implementations and configurations of specialised large language models (LLMs), such as remote-sensing-Qwen2-VL-2B-Instruct-GGUF, which is trained to interpret aerial and satellite imagery, have produced encouraging preliminary results. However, it remains unclear how such AI assistants should be designed for application in schools and to what extent they can effectively support students' learning processes.

Aim and Approach

Against this background, the project "KI:EO – Innovative Ways of Earth Observation for Schools – Use of Artificial Intelligence to Promote Earth Observation Data in the Classroom" aims to integrate LLMs into the existing media network consisting of the adaptive e-learning platform "geo:spektiv"(geospektiv.de) and the web-based tool for analysing EO data "BLIF" (blif.de). A key objective is to develop an AI assistant in the form of a chatbot that supports students in their work with EO data by providing feedback on content, methodology, and procedural issues, thereby helping to maximize learning success (see Fig. 1). In addition, teaching concepts and ready-to-use materials will be developed to ensure the effective implementation of the chatbot. These concepts will also focus on AI-related digital literacy to enable critical reflection on the use of AI.

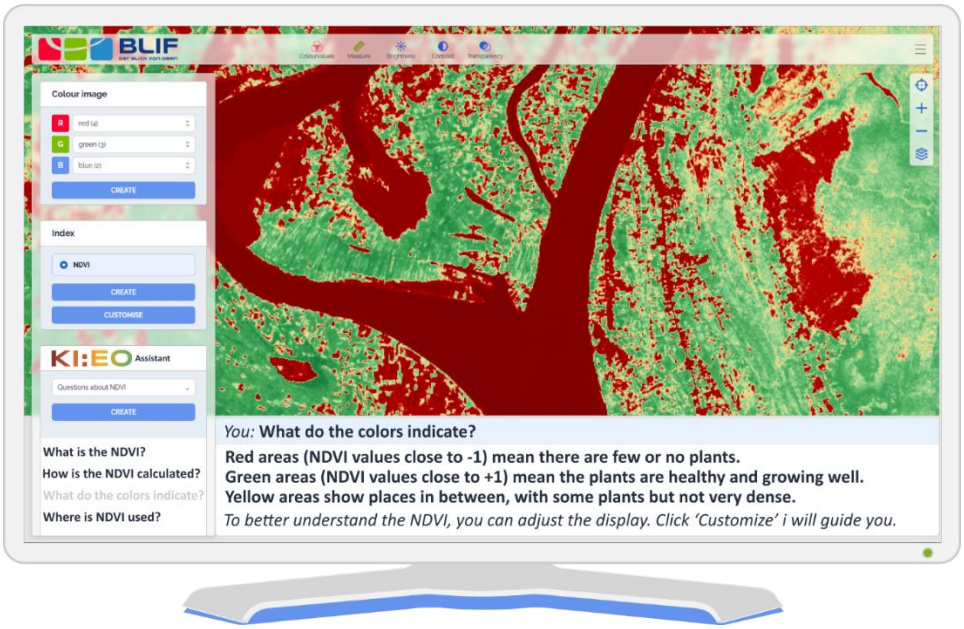

Figure 1. Possible future implementation of the KI:EO chatbot that not only provides pre-written prompts but also suggests useful next actions based on the user's input.

On the technical side, the implementation of AI technology, especially powerful and scalable LLMs, poses significant technical challenges, as sufficient data quality and availability must be ensured. These models need to be trained on accurate educational content and adapted appropriately to prevent inaccurate or misleading responses. Building on analyses of existing AI applications in education and LLMs, particularly those used in EO, a catalogue of functional and non-functional requirements is being developed. These requirements form a central component of the overall teaching concept, which also integrates the extensive experience and proven approaches of the Institute of Geography & Geocommunication as well as concepts from data literacy.

Discussion and Conclusions

AI technologies are widely available and are already being used successfully in education. However, AI-based support for analysing EO data in schools – data that cannot be meaningfully represented in purely textual form – is still in its early stages. It remains to be demonstrated whether AI chatbots can effectively support students' understanding and interpretation of EO data in a school context. Additional key aspects that require discussion include data protection and the role of human oversight in ensuring the quality and accuracy of AI-generated responses. AI may also negatively affect students' thinking and learning if used uncritically or excessively. The AI chatbot to be developed must therefore not undermine critical thinking, problem-solving skills, or creativity. Collaborative learning formats, such as group work and peer discussion, should not be replaced. Similarly, students' motivation for independent effort must be preserved.

Overall, the use of AI to support students holds significant potential. However, its successful implementation requires the integration of a well-designed teaching concept. To achieve the overall goals of the project, a step-by-step implementation of both the AI chatbot and the associated teaching concept, accompanied by continuous evaluation, is essential. The aim of the presentation is to provide initial insights into the overall concept and the planned evaluation process.